

Sentiment Analysis of Reddit Machine Learning Posts

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1 Project Overview and Team

1.1 Team Composition and Roles

Tisha Kotadia (Data Collector & Preprocessor): Responsible for downloading the Reddit ML dataset from Kaggle, performing exploratory data analysis, implementing Reddit-specific text preprocessing (handling markdown, code blocks, URLs, usernames), designing the SQL database schema, and populating the database with processed results and engagement metrics.

Shalavya Agrawal (Model Builder): Focuses on implementing TF-IDF feature extraction with bigrams, building and training machine learning models (Logistic Regression and Naïve Bayes) for sentiment classification, conducting hyperparameter tuning with cross-validation, performing comprehensive model evaluation, and documenting model configurations for reproducibility.

Meghana Narayana (Analyst & Presenter): Interprets model results, queries the database for statistical analysis, creates visualizations including sentiment distributions and time-series trends showing ML community sentiment evolution, analyzes correlations between sentiment and engagement metrics (scores, comments), and prepares the final report and presentation.

1.2 Problem Statement and Motivation

The r/MachineLearning subreddit is one of the largest online communities for ML practitioners and researchers, generating extensive technical discussions about research papers, tools, and methodologies. Sentiment analysis of this community provides insights into how the ML community responds to new techniques, papers, and trends. As CS graduate students specializing in machine learning, analyzing sentiment in domain-specific technical discourse is highly relevant to our field and provides practical experience with real-world NLP applications.

1.3 Goals and Expected Outcomes

Primary Goals: (1) Develop an end-to-end sentiment analysis pipeline from raw Reddit data to actionable insights, (2) Build and compare multiple ML models achieving >75% accuracy, (3) Create time-series visualizations revealing ML community sentiment patterns, (4) Implement SQL database for structured storage of results and metadata, (5) Ensure complete reproducibility through comprehensive documentation.

Expected Outcomes: Functional Logistic Regression and Naïve Bayes classifiers optimized for technical text, clean preprocessing pipeline handling Reddit-specific formatting, relational database with sentiment scores and engagement metrics, interactive Plotly visualizations showing temporal trends, and fully reproducible workflow enabling replication.

1.4 Project Scope

In Scope: Data collection from Kaggle's Reddit ML dataset (120K+ posts), Reddit-specific preprocessing (markdown removal, technical terminology preservation), sentiment labeling based on post scores, TF-IDF feature extraction with bigrams, binary and multi-class sentiment classification, SQL database integration with engagement metrics, time-series trend analysis of ML community sentiment, and statistical visualization.

Out of Scope: Real-time streaming analysis, deep learning models (BERT, transformers), multi-language support, paid APIs, sarcasm detection, and emotion classification beyond basic sentiment categories.

2 Dataset and Methodology

2.1 Dataset Description

Reddit r/MachineLearning Posts Dataset

- **Source:** Kaggle - <https://www.kaggle.com/datasets/fishboi/redditmachinelearning>
- **Size:** 120,000+ posts from r/MachineLearning subreddit
- **Format:** CSV with post ID, title, selftext (body), score, timestamp, author, comments count
- **Content:** ML research discussions, project showcases, questions, and community discourse
- **Accessibility:** Freely available on Kaggle, no authentication required
- **Temporal Coverage:** Multi-year data with Unix timestamps for time-series analysis

- **Advantages:** High-quality technical content, domain relevance to our field, natural sentiment indicators (scores), rich engagement metadata, authentic ML community discourse

2.2 Technical Approach

Phase 1 - Data Collection & Exploration (Weeks 1-2): Download dataset from Kaggle, perform EDA on post structure and content distribution, analyze post scores as sentiment indicators, examine temporal posting patterns, identify data quality issues (missing selftext, deleted posts), and create sentiment labeling strategy based on score thresholds.

Phase 2 - Preprocessing & Database (Weeks 3-4): Combine title and selftext fields for complete content, implement Reddit-specific cleaning (remove markdown, code blocks, URLs, username, subreddit mentions), handle technical ML jargon preservation, apply tokenization and custom stopword removal (excluding domain terms), perform lemmatization, create sentiment labels (positive/neutral/negative based on scores), extract TF-IDF features with bigrams, design normalized SQL schema (posts, sentiments, temporal_data, engagement_metrics tables), and populate database with processed results.

Phase 3 - Model Development (Weeks 5-6): Split data (70% train, 15% validation, 15% test), train Logistic Regression with L2 regularization, train Multinomial Naïve Bayes, perform hyperparameter tuning via grid search with cross-validation, evaluate using accuracy, precision, recall, F1-score, and confusion matrices, select optimal model, save trained models in pickle format.

Phase 4 - Analysis & Visualization (Weeks 7-8): Query database for sentiment distributions by post type, create time-series visualizations showing ML community sentiment trends over months/years, analyze sentiment patterns around major ML conferences (NeurIPS, ICML), generate word clouds for each sentiment class, analyze feature importance to identify key terms, investigate correlation between sentiment and engagement metrics, and create interactive Plotly dashboards.

Phase 5 - Documentation (Week 9): Draft final report with methodology and results, create presentation slides, update README with reproduction instructions, test reproducibility in clean environment, and prepare final submission package.

2.3 Technology Stack

Programming Language: Python 3.9+ (team expertise, rich NLP/ML ecosystem)

Core Libraries: pandas/numpy (data processing), NLTK (NLP - tokenization, stopwords, lemmatization), scikit-learn (ML models, TF-IDF, evaluation), SQLite with SQLAlchemy (database), matplotlib/seaborn/plotly (visualization), wordcloud (word clouds), Jupyter Notebook (development)

Version Control: Git/GitHub for collaborative development

Documentation: Overleaf (LaTeX) for reports, Markdown for technical docs

Rationale: Selected based on team proficiency, industry standards, and specific requirements for NLP sentiment analysis. All tools are open-source and freely available.

3 Implementation Plan and Deliverables

3.1 Task Distribution

Tisha Kotadia: Download and verify Reddit ML dataset, conduct EDA, implement Reddit-specific preprocessing pipeline, design sentiment labeling strategy, create SQL database schema with engagement tables, develop data loading scripts, document preprocessing workflow.

Shalavya Agrawal: Implement TF-IDF with bigrams, train and optimize Logistic Regression, train and optimize Naïve Bayes, conduct hyperparameter tuning and cross-validation, compare model performance, save trained models, document model configurations and results.

Meghana Narayana: Write SQL queries for analysis, create sentiment distribution visualizations, develop time-series trend plots, generate word clouds, analyze engagement correlations, investigate sentiment patterns around ML events, interpret results, prepare final report and presentation.

3.2 Project Timeline and Milestones

Milestone	Week	Owner	Deliverable
Dataset Downloaded	1	Tisha	Reddit ML CSV
EDA Complete	2	Tisha	Analysis notebook
Preprocessing Done	3	Tisha	Clean data scripts
Database Populated	4	Tisha	SQL database
Models Trained	5-6	Shalavya	Trained models
Visualizations Ready	7-8	Meghana	Charts & plots
Final Submission	9	All	Complete package

3.3 Key Deliverables by Phase

Phase 1: Downloaded dataset, initial EDA report, data quality assessment, sentiment labeling strategy

Phase 2: Preprocessing pipeline (Jupyter notebook), SQL database with schema, cleaned data loaded, data quality report

Phase 3: Two trained ML models, model evaluation metrics, hyperparameter tuning results, comparison analysis, saved models

Phase 4: Time-series sentiment visualizations, word clouds, engagement correlation analysis, statistical reports, interactive dashboard

Phase 5: Final IEEE-format report, presentation slides, reproducibility documentation, complete GitHub repository

3.4 Development Environment and Setup

Requirements: Python 3.9+, 8GB RAM, 5GB disk space, stable internet

Installation: (1) Clone GitHub repo, (2) Create virtual environment (`python -m venv sentiment-env`), (3) Install dependencies (`pip install -r requirements.txt`), (4) Download NLTK data (`nltk.download()`), (5) Verify installation

Repository Structure:

```
sentiment-analysis-social-media/  
data/raw/           # Reddit ML dataset  
notebooks/          # 4 Jupyter notebooks (EDA, preprocessing,  
                    # modeling, visualization)  
src/                # Python modules (preprocessing.py,  
                    # models.py, etc.)  
database/           # SQLite database  
models/             # Saved trained models  
reports/            # PDF report, figures  
progress_tracker.xlsx
```

Version Control: Main branch (stable), dev branch (active work), feature branches per member. All code changes require peer review before merging.

GitHub Repository: <https://github.com/Shalavya8103/Social-Media-Sentiment-Analysis>

3.5 Skills Assessment and Risk Mitigation

Team Strengths: Advanced Python programming, ML fundamentals from coursework, data preprocessing experience, SQL proficiency, visualization skills, Git expertise.

Learning Needs: NLP libraries (NLTK) - Week 1 team tutorial planned, Reddit data handling - documentation review, time-series visualization (Plotly) - guided learning, database-ML integration (SQLAlchemy) - code examples provided.

Technical Risks & Mitigation: (1) Reddit markdown complexity → Custom preprocessing functions with thorough testing, (2) Technical jargon in ML posts → Custom stopwords list preserving domain terms, (3) Imbalanced sentiment distribution → Stratified sampling if needed, (4) Model performance issues → Multiple algorithms and extensive tuning.

Timeline Risks & Mitigation: (1) Preprocessing delays → Start with data subset, parallelize work, (2) Team member unavailability → Cross-training on components, clear documentation.

External Resources: Professor office hours, course TAs, Stack Overflow community, NLTK/sklearn documentation, IEEE papers on sentiment analysis.

3.6 Success Criteria

Project success defined by: (1) Complete reproducible pipeline from raw data to results, (2) ML models achieving >75% test accuracy, (3) Functional SQL database with proper schema, (4) Clear time-series visualizations of ML community trends, (5) Comprehensive documentation enabling full replication, (6) On-time milestone delivery, (7) Insights into ML community sentiment patterns.

Progress Tracking: All tasks tracked in Excel progress tracker (included in submission) with weekly updates during team meetings. Tracker includes task IDs, task name, team members, and phases.

Dataset Link: <https://www.kaggle.com/datasets/fishboi/redditmachinelearning>